Also, we capped the episode duration at 200, so if you increase the cap it will be able to play even longer.

4.4.5. Chapter conclusion

REINFORCE is an effective and very simple way of training a policy function, but it's a little too simple. For CartPole it works very well, since the state space is very small and there are only two actions. If we're dealing with an environment with many more possible actions, reinforcing all of them each episode and hoping that on average it will only reinforce the good actions becomes less and less reliable. In the next two chapters we'll explore more sophisticated ways of training the agent.

Summary

- *Probability* is a way of assigning degrees of belief about different possible outcomes in an unpredictable process. Each possible outcome is assigned a probability in the interval [0,1] such that all probabilities for all outcomes sum to 1. If we believe a particular outcome is more likely than another, we assign it a higher probability. If we receive new information, we can change our assignments of probabilities.
- *Probability distribution* is the full characterization of assigned probabilities to possible outcomes. A probability distribution can be thought of as a function $P:O \rightarrow [0,1]$ that maps all possible outcomes to a real number in the interval [0,1] such that the sum of this function over all outcomes is 1.
- A *degenerate probability distribution* is a probability distribution in which only 1 outcome is possible (i.e., it has probability of 1, and all other outcomes have a probability of 0).
- *Conditional probability* is the probability assigned to an outcome, assuming you have some additional information (the information that is conditioned).
- A *policy* is a function, $\pi S \rightarrow A$, that maps states to actions and is usually implemented as a probabilistic function, $\pi P(A|S)$, that creates a probability distribution over actions given a state.